

COMPANION TO THE TEAM CONTRACT

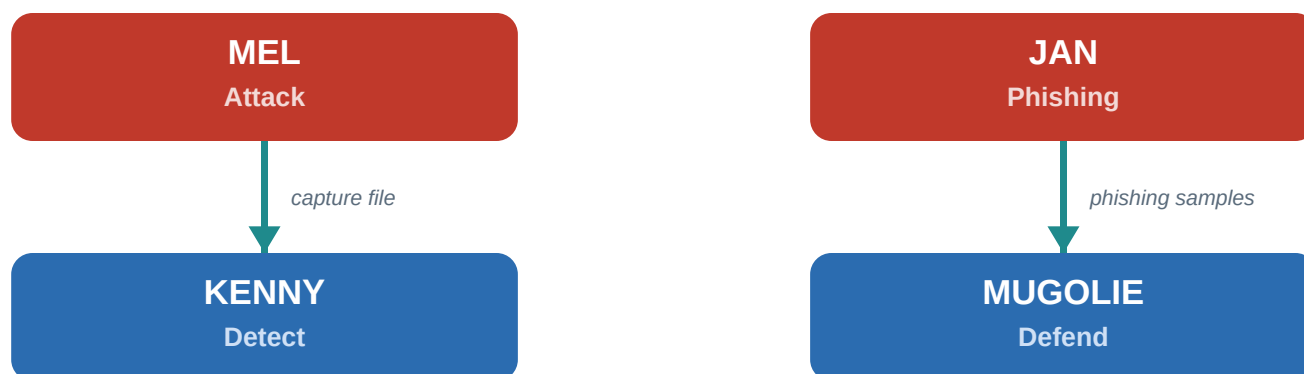
Operation Catch the Hacker: Step by Step Playbooks

One simple guide per person. Find your name, do your part.

How to use this. Each person has their own page with their own steps. Find your name, read only your page, and follow it top to bottom. The dark boxes are things you type exactly as written. Anything in pointy brackets like <target-ip> is something you fill in with your own value.

How the Four Parts Connect

RED TEAM makes it happen, BLUE TEAM responds



Mugolie also uses Mel's attack to test the firewall.

Mel attacks and saves a recording, which Kenny opens to catch the attack. Jan makes phishing samples, which Mugolie turns into a how-to-spot-it guide. Mugolie also locks down the machine so Mel's attack stops working. Everything is posted to your team Slack channel, so you can each work at a different time.

Before Anyone Starts: Install These

Set these up once. They are all free. If you only need part of this for your role, your page will tell you.

Tool	What it is
VirtualBox or Parallels	Runs other computers inside yours. Parallels on the MacBooks, VirtualBox on the Lenovo and class Windows.
Two Kali Linux VMs	One is your attacker, one is the target. Kali already has Nmap, Hydra, Wireshark, and Snort built in.
AI assistant	A free Claude account, used to explain alerts and to write phishing samples to study.
Team Slack channel	Where every file and write-up gets posted. Save written pieces as PDFs.

Good news on hardware. Because the attacker and the target are both Kali, the whole project runs on the MacBooks in Parallels, on Kenny's Lenovo, and on the class Windows machines. Every tool is already inside Kali, so there is nothing extra to download.

WHAT YOU DO

Set up a target machine, scan it, then break in by guessing a weak password. Record the whole thing so Kenny can study it. You work on your own and record a short walkthrough for the team.

TOOLS YOU USE

Tool	What it is
Nmap	Lists the open doors (services) on the target.
Hydra	Tries many passwords fast to guess a login.
Wireshark	Records all the network traffic as proof of the attack.

YOUR STEPS

- Start two Kali VMs: one is your attacker, one is your target. On your MacBook use Parallels. They just need to be on the same network so they can see each other.
- Set up the target. On the target Kali, turn on SSH, give the kali user a simple password like password123, and note its IP address (next to eth0).

```
sudo systemctl enable ssh --now
sudo passwd kali
ip a
```

- On the attacker Kali, open Wireshark, click your adapter (usually eth0), and click the blue shark-fin to start recording.
- Scan the target. You should see SSH open on port 22.

```
nmap -sV <target-ip>
```

- Make a small password list. Add a few common passwords including the one you set, one per line, then save and close.

```
nano passwords.txt
```

- Guess the login with Hydra. When it finds the password, you are in.

```
hydra -l kali -P passwords.txt ssh://<target-ip>
```

- Stop the Wireshark recording (red square) and save it as capture.pcap.
- Take screenshots of the scan, Hydra running, and the password it found.
- Record a short screen video so the team can present your part while you are away.

POST TO SLACK

- capture.pcap, your screenshots, a short write-up saved as a PDF, and your walkthrough video or a link to it. Post it all in the team Slack channel.

YOU ARE DONE WHEN

- Hydra found the password and capture.pcap is posted in Slack.

WHAT YOU DO

Study how attackers gather information before a phishing attack, then use the AI assistant to write sample phishing emails so the team can learn the warning signs. No real companies or real people, only our practice company.

TOOLS YOU USE

Tool	What it is
AI assistant	Writes example phishing emails for you to study.
A doc or PDF	Where you write your notes and samples, then export to a PDF.

YOUR STEPS

- Write a short note titled The Attacker's View. List the kinds of information a scammer looks for before phishing a company: employee names, the email format (like first.last@company), job titles, and what software the company uses. Use our practice company, Unicorns Wireless Corp, as the made-up example.
- Open the AI assistant and ask it: write three example phishing emails aimed at a small company so my team can study the warning signs.
- For each email, list the red flags. Look for a fake sender address, an urgent threat, a strange link, and bad spelling.
- Save the three samples and the red flags as one PDF, and tell Mugolie in Slack that it is ready for the phishing guide.

POST TO SLACK

- Your Attacker's View note and the three phishing samples with their red flags, saved as one PDF and posted to Slack.

YOU ARE DONE WHEN

- You have three phishing samples with red flags, ready for Mugolie.

Tip. You can do this part any time, on any computer. You do not need the VMs. If the group prefers, you can join the Blue Team instead, as long as all four roles stay covered.

WHAT YOU DO

Open Mel's recording, find the attack inside it, and write simple rules that would have caught it. Then use the AI assistant to explain the alerts in plain English.

TOOLS YOU USE

Tool	What it is
Wireshark	Opens the recording and filters for the attack traffic.
Snort	An alarm system. You write rules and it shouts when it sees a match.
AI assistant	Explains a confusing alert like a human teammate would.

YOUR STEPS

- Get capture.pcap from Mel's post in Slack and open it in Wireshark.
- Go to Statistics, then Conversations. The computer that did the most talking is the attacker. Note its IP.
- Find the scan. Paste this filter in the bar at the top. Lots of these to many ports means a scan.

```
tcp.flags.syn==1 && tcp.flags.ack==0
```

- Find the break-in. Change the filter to **tcp.port==22** to see the many SSH login attempts from the password guessing.
- Make a file called local.rules with the one line below in it.

```
alert tcp any any -> any 22 (msg:"SSH login attempts, possible brute force";  
sid:1000001; rev:1;)
```

- Run Snort against the recording. It prints an alert each time the rule matches.

```
sudo snort -r capture.pcap -c local.rules -A console
```

- Copy one alert, paste it into the AI assistant, and ask: explain this alert in plain English and tell me what an analyst should do. Use the answer to write your findings.

POST TO SLACK

- Your local.rules file, screenshots of the Snort alerts, and a short plain-English findings write-up saved as a PDF. Post them to Slack.

YOU ARE DONE WHEN

- Snort prints alerts for the attack and your findings explain it in plain words.

WHAT YOU DO

Turn on a firewall so Mel's attack fails, prove it with a before and after, then turn Jan's phishing samples into a one-page guide on spotting phishing.

TOOLS YOU USE

Tool	What it is
UFW firewall	A simple firewall built into Linux. You turn it on and add block rules.
Nmap	Used here to prove the doors are now closed.
Jan's phishing samples	The raw material for your spot-the-phishing guide.

YOUR STEPS

- On the target machine, turn the firewall on.

```
sudo ufw enable
```

- Block the attacker's IP and close the SSH door, then check it.

```
sudo ufw deny from <attacker-ip>
sudo ufw deny 22/tcp
sudo ufw status
```

- From Kali, run the same scan Mel used: **nmap -sV <target-ip>**. Port 22 should now show as filtered or closed. Take a before and after screenshot.
- Get Jan's phishing samples from his post in Slack. Write a one-page guide titled How to Spot a Phishing Email, using the red flags from those samples.

POST TO SLACK

- Your before and after scan screenshots and your one-page How to Spot a Phishing Email guide, saved as a PDF. Post them to Slack.

YOU ARE DONE WHEN

- The attack is blocked in the after screenshot and your phishing guide is finished.

Four playbooks, one team. Find your name, follow your steps, and we catch the hacker together.